

How Inheritance Works

To create a new child class, you put the name of the existing parent class in between parentheses in the `class` statement. To practice creating a child class, open a new file editor window and enter the following code; save it as *inheritanceExample.py*:

```
❶ class ParentClass:
❷     def printHello(self):
        print('Hello, world!')

❸ class ChildClass(ParentClass):
        def someNewMethod(self):
            print('ParentClass objects don't have this method.')

❹ class GrandchildClass(ChildClass):
        def anotherNewMethod(self):
            print('Only GrandchildClass objects have this method.')

print('Create a ParentClass object and call its methods:')
parent = ParentClass()
parent.printHello()

print('Create a ChildClass object and call its methods:')
child = ChildClass()
child.printHello()
child.someNewMethod()

print('Create a GrandchildClass object and call its methods:')
grandchild = GrandchildClass()
grandchild.printHello()
grandchild.someNewMethod()
grandchild.anotherNewMethod()

print('An error:')
parent.someNewMethod()
```

When you run this program, the output should look like this:

```
Create a ParentClass object and call its methods:
Hello, world!
Create a ChildClass object and call its methods:
Hello, world!
```